

Junior Automation & Controls Technician

Four University-Structured Industrial Simulation Project Briefs

PROGRAMME INFORMATION	
PROGRAMME WINDOW	7 September 2026 - 28 February 2027
PORTFOLIO	Exactly four substantial, progressively advanced projects
DELIVERY MODEL	Software simulation only - no physical construction required
TARGET ROLES	Junior automation, controls, PLC, instrumentation and E&I technician

PURPOSE

Demonstrate industrial reasoning through requirements, PLC engineering, process simulation, HMI/SCADA, networking, structured testing, troubleshooting, commissioning and traceable documentation.

Prepared for Emmanuel Januarie

College-style personal learning programme - not an accredited qualification

Portfolio purpose and progression

These projects are assessment briefs for entry-level Junior Automation Technician, Junior Controls Technician, Automation & Controls Technician, Instrumentation & Controls Technician, PLC Technician, E&I Technician and related roles. They assess applied ability rather than the ability to copy a tutorial.

Project	Progression	Syllabus relationship
1 - Motor & Conveyor Control	Electrical control -> PLC I/O -> Ladder Logic -> motor/process sequence	M1 Electrical Fundamentals + M2 PLC & Automation
2 - Process & Instrumentation	Analog measurement -> scaling -> sequencing -> process control -> PID	M3 Instrumentation + M4 Process Control
3 - SCADA & Communications	PLC data -> HMI/SCADA -> alarms -> historian -> industrial protocols	M5 SCADA/HMI + M6 Industrial Networking
4 - Integrated Remote Process Package	Documentation -> integration -> commissioning -> structured fault-finding	M7 Documentation + M8 Safety/Commissioning, integrating M1-M6

PROGRESSIVE STANDARD

Each project must reuse concepts and selected software artifacts from earlier work. Project 4 is the flagship integration and must not be four unrelated demonstrations placed in one folder.

University-brief principle

The document specifies the problem, required functions, constraints, evidence and marking criteria. It intentionally does not prescribe rung-by-rung logic, screen layouts, tag names, database schemas or fault solutions. Those engineering decisions are part of the assessment.

Simulation environment and approved software

No physical components are required. Motors, pumps, valves, switches, transmitters, analog loops, process dynamics and communication failures are represented through software I/O, process models and controlled fault injection.

Capability	Preferred industrial platform	Acceptable equivalent / fallback
PLC engineering	Siemens TIA Portal + S7-PLCSIM for S7-1200/S7-1500	CODESYS Control or another IEC 61131-3 environment; record differences
Process/machine simulation	Factory I/O connected to the PLC simulator	A documented dynamic process simulator or custom signal model with credible behavior
HMI/SCADA	Ignition Perspective / Maker Edition for personal study	WinCC, AVEVA, FactoryTalk View, CODESYS Visualization or equivalent if legitimately available
Historian/data	Ignition Historian or SQL Historian	A documented SQL/time-series store linked to the SCADA model
Industrial protocols	OPC UA and Modbus TCP/RTU simulation	Platform-native simulator or protocol server/client with register/tag map
Network diagnostics	Wireshark plus operating-system IP tools	Equivalent packet/connection diagnostics with saved evidence
Engineering documents	EPLAN/AutoCAD Electrical where accessible	QElectroTech, diagrams.net or another controlled drawing tool

Software-selection rules

- [] Confirm version, licence and connector compatibility before committing to the architecture.
- [] Record the exact software versions, drivers, protocol endpoints and limitations used.
- [] Use one primary PLC platform consistently unless vendor translation is an explicit task.
- [] Do not claim a platform connection, protocol or test that was not demonstrated.
- [] If a preferred product is unavailable, preserve the engineering requirement and document the substitute.

VERIFIED CAPABILITY BASIS

Current Siemens S7-PLCSIM supports debugging and validation without PLC hardware. Factory I/O documents connections to S7-PLCSIM. Ignition Maker provides personal, non-commercial access to Perspective, OPC UA and historian-related capabilities. Wireshark supports capture and display-filter analysis.

Industrial simulation and evidence standards

REQUIRED SIMULATION QUALITIES

- ☐ Defined scan/update behavior
- ☐ Engineering-unit ranges
- ☐ Command and feedback separated
- ☐ Manual/Automatic ownership
- ☐ Permissives and interlocks
- ☐ Alarm/trip recovery rules
- ☐ Bad/stale data quality
- ☐ Repeatable injected faults

PROHIBITED CLAIMS

- ☐ No live-plant commissioning claim
- ☐ No certified safety validation
- ☐ No hazardous-area competence claim
- ☐ No real offshore experience claim
- ☐ No hardware installation claim
- ☐ No protocol claim without evidence
- ☐ No fabricated screenshots
- ☐ No hidden unperformed tests

Evidence labels

Label	Meaning
SIMULATED	The behavior was executed in software with defined inputs, outputs and model assumptions.
IMPLEMENTED	The configuration, logic, display, communications or database artifact exists and is reviewable.
TESTED	Expected and actual results, evidence, defects and retest status were recorded.
DOCUMENTED	A revision-controlled artifact is included and coordinated with related documents.
CONCEPT ONLY	The subject is explained but not configured or executed.
NOT TESTED	The requirement or condition was not performed and is excluded from the claim.

SAFETY REPRESENTATION

E-stop, ESD, trips, permissives and hazardous-area boundaries are functional simulations for learning. They are not safety-rated designs and must never be presented as certified protective systems.

Common project instructions

Apply this lifecycle to every project. The sequence defines required engineering stages, not a software walkthrough.

Stage	Required decision or output
1 - Interpret	Extract process intent, users, modes, constraints, exclusions and measurable success criteria.
2 - Specify	Create functional requirements, I/O, tags, signal ranges, states, alarms and interfaces.
3 - Design	Select architecture; define control philosophy, sequences, ownership and protective behavior.
4 - Implement	Configure the PLC, process model, HMI/SCADA, communications and data functions.
5 - Verify	Trace every requirement to normal, boundary, fault and recovery tests.
6 - Diagnose	Introduce controlled faults; preserve symptom, evidence, isolation, cause and verification.
7 - Release	Resolve defects, run regression, update as-built documents, back up and demonstrate.

Required folder pattern

PORTFOLIO STRUCTURE

00_README | 01_Brief_and_Requirements | 02_Design | 03_PLC | 04_Process_Model | 05_HMI_SCADA | 06_Network_Data | 07_Testing | 08_Faults | 09_Demo | 10_Backups

Minimum evidence index fields

- [] Evidence ID, project, title, revision and date
- [] Software/platform/version and simulation label
- [] Requirement, interface or test proved
- [] Expected result, actual result and pass/fail state
- [] Known limitation, unresolved defect or NOT TESTED statement
- [] File path, screenshot/source relationship and backup location

Common assessment model

Each project is marked out of 100. Project-specific rubrics refine these categories without changing the expectation of traceability and independent engineering judgment.

Category	Weight	Common evidence standard
Requirements and scope	10%	Requirements are measurable; assumptions, exclusions and limitations are explicit.
Architecture and design	15%	Interfaces, ownership, data paths, states and protective intent are coherent.
Technical implementation	25%	PLC/process/HMI/network behavior is correct, readable and maintainable.
Fault handling and diagnostics	15%	Abnormal states are detected, explained, protected and recoverable.
Verification and evidence	20%	Tests trace to requirements and include expected/actual results and regression.
Engineering documentation	10%	Documents agree with configuration, logic, tags and tests.
Technical defence	5%	The learner explains decisions, tradeoffs, faults and limitations independently.

Grade descriptors

Band	Descriptor
85-100 Distinction	Integrated, traceable and professionally presented; fault reasoning and limitations are unusually strong.
75-84 Competent	Requirements are met with sound implementation, testing and documentation; minor gaps do not undermine claims.
60-74 Developing	Core operation works, but integration, diagnostics, evidence or document consistency needs improvement.
Below 60 Not released	Important functions are incomplete, unproved or poorly understood.

SAFETY-CRITICAL RELEASE RULE

Any unresolved simulated E-stop/ESD, trip, restart, command-ownership or bad-quality-data defect prevents release regardless of total score.

Project 1 - Industrial Motor & Conveyor Control Simulation

RECOMMENDED WINDOW	SYLLABUS ALIGNMENT	SUBMISSION TYPE
5 October - 15 November 2026	M1 Electrical Fundamentals + M2 PLC & Automation	Individual industrial simulation project

PROJECT SCENARIO

A distribution facility requires a two-zone conveyor controller that moves items from an infeed point to a discharge point. The simulated system must provide safe starting, clear command ownership, object tracking, controlled stopping and diagnosable faults.

Assessment purpose

Establish the industrial chain: electrical/control intent -> field I/O -> PLC scan and Ladder Logic -> output command -> simulated motor/process response.

Required software

Function	Preferred tool	Evidence expected
PLC	TIA Portal + S7-PLCSIM	Hardware configuration, tag table, blocks, monitoring and diagnostics
Machine/process	Factory I/O conveyor scene or equivalent	Mapped sensors/actuators and repeatable object behavior
Operator interface	Ignition Perspective or equivalent HMI	Overview, modes, commands, status, faults and count
Documentation	Controlled drawing/document tool	Schematic, I/O list, narrative, sequence and tests

Learning outcomes

- [] Translate an operating description into I/O, Ladder Logic, modes and tests.
- [] Apply start/stop, seal-in, timers, counters, one-shots, permissives and interlocks.
- [] Separate commands, outputs and simulated running feedback.
- [] Diagnose electrical-control, I/O, logic and sequence faults by layer.

Project 1 - functional specification

Develop the implementation independently. Your solution must meet the following outcomes; the brief does not prescribe rung structure or tag naming.

Required functions

- [] Two conveyor zones or equivalent motor sections with independent command and feedback states.
- [] Start, Stop, Reset and training E-stop/healthy-chain simulation.
- [] Local/Remote and Manual/Automatic command ownership with defined transition behavior.
- [] Start permissives, motor interlocks and restart inhibition after a trip or simulated power cycle.
- [] Entry, transfer and exit detection with one count per item.
- [] Start-failure, transfer timeout/jam, sensor discrepancy and overload/VFD-fault representation.
- [] Alarm acknowledgement separated from fault correction and reset.
- [] HMI indication of command, feedback, mode, owner, permissives, interlocks and first relevant fault.

Engineering constraints

Constraint	Requirement
Control authority	The PLC retains sequencing and protection; the HMI requests actions but does not replace PLC logic.
Feedback	Motor running feedback must not be derived directly from the run command.
Restart	No automatic restart after training E-stop, overload/trip or simulated power restoration.
Counting	A maintained or chattering sensor must not increment the counter repeatedly.
Traceability	Every I/O, tag, alarm and test uses a consistent identifier.

CHALLENGE

Objects arrive at irregular intervals. Design sequence and count logic that remains correct when sensors stay active across several PLC scans and when an item stalls between zones.

Project 1 - required work and submission

ENGINEERING WORK	SUBMISSION PACKAGE
☐ Define assumptions and exclusions	☐ Requirements specification
☐ Create architecture and I/O list	☐ Electrical/control schematic
☐ Write control narrative and sequence	☐ PLC I/O and tag lists
☐ Develop commented PLC logic	☐ Control narrative
☐ Configure the conveyor/process model	☐ Sequence/state diagram
☐ Create the operator interface	☐ PLC source/export and screenshots
☐ Design normal/fault/recovery tests	☐ HMI screens
☐ Resolve defects and issue as-built files	☐ Test and fault reports
	☐ Evidence index and 5-minute demonstration

Mandatory fault campaign

Fault	Required evidence
Sensor stuck on/off	Symptom, I/O/tag state, sequence effect, diagnosis and verified recovery
Motor command without feedback	Start-failure timing, alarm/trip behavior and reset conditions
Missing permissive	Blocked start plus operator-visible cause
Transfer jam	Timeout, stopped outputs, retained diagnostic cause and controlled restart
HMI request in Local	Command rejection and ownership indication
Logic defect	Change record, corrected logic and regression of unaffected functions

Project 1 - verification and marking

Project-specific marking rubric

Category	Weight	What earns strong marks
Requirements/I/O design	15%	Complete, consistent I/O, modes, sequence and measurable acceptance criteria.
PLC implementation	25%	Readable Ladder Logic correctly handles scan behavior, timers, counters and state.
Modes/protection	20%	Ownership, permissives, interlocks, trips and restart behavior are explicit.
Simulation/HMI	15%	Process behavior and operator information agree with PLC states.
Fault tests	15%	Six faults are diagnosed and recovered using evidence.
Documentation/defence	10%	Documents agree and the learner explains decisions without a walkthrough.

RELEASE STANDARD

Minimum 75/100, all mandatory functions demonstrated, all six faults closed, and no unresolved restart, ownership or protective-response defect.

Examiner questions

- ☐ Why is running feedback independent of the command?
- ☐ How does the design prevent one object from being counted repeatedly?
- ☐ What prevents a restart after a trip?
- ☐ Where would you investigate if the process sensor is true but the PLC tag is false?
- ☐ Which project claim remains simulated rather than industrially validated?

Project 2 - Automated Process & Instrumentation Simulation

RECOMMENDED WINDOW	SYLLABUS ALIGNMENT	SUBMISSION TYPE
16 November - 27 December 2026	M3 Instrumentation + M4 Process Control, extending Project 1 PLC skills	Individual industrial simulation project

PROJECT SCENARIO

A process package transfers liquid from a feed reservoir through duty/standby pumps into a process vessel, then through a controlled outlet to storage. Instrumentation, sequencing and protective functions must keep the simulated process within defined operating limits.

Assessment purpose

Establish the chain: process -> instrument -> signal -> PLC scaling/control -> final element -> process response.

Required software

Function	Preferred tool	Evidence expected
PLC	TIA Portal + S7-PLCSIM	Analog/digital configuration, sequencing, scaling, alarms and PID fundamentals
Process model	Factory I/O fluid scene or equivalent dynamic model	Level/flow/pressure/temperature response and disturbances
HMI	Ignition Perspective or equivalent	P&ID-style overview, faceplates, alarms and trends
Signals	Simulator tags/math representing 4-20 mA and failures	Raw value, current, percent, engineering value and quality

Learning outcomes

- [] Select and document simulated measurement ranges and signal types.
- [] Scale and validate analog values, including under/over-range and bad quality.
- [] Develop pump/valve sequencing, Manual/Automatic control and protective responses.
- [] Interpret PV/SP/MV and PID response without confusing process faults with instrument faults.

Project 2 - functional specification

The process model must be dynamic enough for operator action, controller output and disturbances to produce visible trends.

Required measurements and control

- ☐ Continuous vessel level plus independent low-low and high-high protective inputs.
- ☐ Temperature measurement with defined range, units, normal band and alarm limits.
- ☐ Flow measurement or credible calculated flow with a no-flow condition.
- ☐ Pressure measurement concept using a simulated 4-20 mA transmitter.
- ☐ Raw-to-engineering scaling and reverse calculation for at least three analog signals.
- ☐ Duty/standby pump selection, automatic failover and unavailable-device handling.
- ☐ Outlet valve command/position discrepancy and defined fail-state assumption.
- ☐ Level control using PID fundamentals or a justified alternative control strategy.
- ☐ Alarm, interlock, permissive and trip behavior coordinated through cause/effect.

Calibration and loop requirements

Activity	Minimum evidence
Three-point verification	Low, midpoint and high input with expected/actual raw, mA, percent and engineering value
As-found/as-left	One intentionally incorrect range or bias, identified and corrected without hiding the original result
Loop check	Process value -> transmitter model -> PLC raw/scaled tag -> HMI display/alarm
Plausibility	Detection or documented handling for frozen, noisy, under-range and over-range values

CHALLENGE

Maintain vessel level through a changing outlet disturbance while one duty pump becomes unavailable. The design must avoid high-high level, indicate degraded operation and recover without an uncontrolled output step.

Project 2 - required work and submission

ENGINEERING WORK	SUBMISSION PACKAGE
☐ Define process and ranges	☐ Requirements and process assumptions
☐ Create P&ID and instrument list	☐ P&ID and control schematic
☐ Develop I/O and loop architecture	☐ Instrument/I/O/tag lists
☐ Write sequence/control narrative	☐ Loop diagrams
☐ Configure scaling and process model	☐ Control narrative and sequence
☐ Develop PLC and HMI functions	☐ Alarm list and cause/effect
☐ Perform calibration/loop checks	☐ Calibration and loop sheets
☐ Run faults, regression and release	☐ PLC/HMI/process-model files
	☐ Test/fault reports and demonstration

Mandatory fault campaign

Fault	Required evidence
Incorrect scaling	As-found value, calculation, configuration cause, correction and three-point retest
Frozen/noisy level	Trend evidence, quality/plausibility response and safe process action
Duty pump failure	Failover behavior, alarm, capacity limitation and recovery
Valve fails to move	Command/feedback discrepancy and process consequence
No-flow condition	Pump command, flow response, delay, trip/alarm and diagnosis
High-high level	Independent protective response and controlled restoration

Project 2 - verification and marking

Project-specific marking rubric

Category	Weight	What earns strong marks
Process/instrument design	15%	P&ID, ranges, signal types, loops and assumptions form a coherent process.
PLC/process control	25%	Sequences, scaling, modes, pump/valve logic and PID fundamentals behave correctly.
Protection/alarms	15%	Permissives, interlocks, alarms, trips and recovery are coordinated.
Calibration/loop evidence	15%	Raw-to-display traceability and as-found/as-left evidence are correct.
Fault tests	20%	Instrument and process faults are distinguished using trends and layered evidence.
Documentation/defence	10%	The package is coordinated and engineering choices are independently defended.

RELEASE STANDARD

Minimum 75/100, all three analog chains verified, all six mandatory faults closed, and no unresolved high-high, failover, scaling or recovery defect.

Examiner questions

- ☐ How do you distinguish a true process change from a failed instrument?
- ☐ Why is 4 mA a useful live zero?
- ☐ What prevents a controller from hiding a high-high protection requirement?
- ☐ How was the chosen process model validated as credible enough for this project?
- ☐ What evidence proves that the displayed engineering value is correct?

Project 3 - SCADA, Historian & Communications Simulation

RECOMMENDED WINDOW	SYLLABUS ALIGNMENT	SUBMISSION TYPE
28 December 2026 - 24 January 2027	M5 SCADA/HMI + M6 Industrial Networking, supervising Projects 1 and 2	Individual industrial simulation project

PROJECT SCENARIO

A central supervisory system must monitor the conveyor and process package as two areas. It must provide clear operator navigation, alarm management, historical context and diagnosable communications while control authority remains in the PLC simulations.

Assessment purpose

Prove that PLC data can be transformed into reliable operator information and historical evidence without concealing quality, ownership or communication failures.

Required software

Function	Preferred tool	Evidence expected
SCADA/HMI	Ignition Perspective / Maker Edition	Gateway, tag providers, views, faceplates, alarms, trends and security awareness
Controller/process	Project 1 and 2 PLC/process simulations	At least two supervised areas with reusable tag structures
Protocols	OPC UA plus Modbus TCP/RTU simulation	Endpoint/configuration and controlled mapping evidence
Historian	Ignition Historian or SQL Historian	Stored data, retrieval, time range and quality-aware trend evidence
Diagnostics	Wireshark and OS network tools	Filtered capture/connection evidence and layered diagnosis

Learning outcomes

- [] Design usable overview, detail, faceplate, alarm and trend navigation.
- [] Configure meaningful alarm, historian and data-quality behavior.
- [] Document Ethernet, IP, TCP, OPC UA, Modbus and serial concepts in the implemented context.
- [] Diagnose communication failures without immediately blaming PLC logic.

Project 3 - functional specification

The SCADA project must supervise the prior simulations rather than becoming an unrelated dashboard.

Required supervisory functions

- [] Site overview showing both areas, overall health, active alarms and communication state.
- [] Area and equipment views with command, mode, owner, feedback, permissive and fault information.
- [] PLC-enforced command authority; commands disabled or rejected when ownership/conditions are invalid.
- [] Alarm priorities, meaningful messages, acknowledgement, return-to-normal and operator response guidance.
- [] Trends for selected process variables, setpoints, outputs, states and fault events.
- [] Historical storage and retrieval with documented sampling/deadband assumptions.
- [] Bad/stale quality visualization that cannot be mistaken for a healthy live value.
- [] At least one OPC UA connection and one Modbus mapping or documented equivalent.

Network and data requirements

Artifact	Required content
Network diagram	Nodes, interfaces, IP/subnet, protocols, ports, trust/security boundary and failure impact
Address schedule	Unique addresses, role, owner, gateway requirement and status
Protocol map	Tag/register, direction, datatype, scale, byte/word order, update rate and quality
Historian definition	Tags, sample/deadband, retention assumption, timestamp source and retrieval test

CHALLENGE

A communication outage must make current values visibly bad/stale while preserving historical context. When communications recover, command authority and data quality must recover in a controlled, observable manner.

Project 3 - required work and submission

ENGINEERING WORK	SUBMISSION PACKAGE
☐ Define operator tasks and hierarchy	☐ SCADA requirements and style guide
☐ Create tag/alarm conventions	☐ Screen hierarchy/wireframes
☐ Configure PLC/protocol connections	☐ Tag and alarm registers
☐ Develop reusable faceplates/views	☐ Network/address/protocol maps
☐ Configure alarm and historian functions	☐ Historian configuration record
☐ Create network/protocol documents	☐ HMI/SCADA export and screenshots
☐ Inject communications/data faults	☐ Packet/diagnostic evidence
☐ Audit usability and release	☐ Communication fault reports
	☐ Usability review and demonstration

Mandatory fault campaign

Fault	Required evidence
PLC connection loss	Quality change, screen behavior, alarm and controlled recovery
Wrong IP/subnet	Layered checks and corrected address schedule
Wrong Modbus mapping	Register/datatype/byte-order evidence and verified value
Stale data	Timestamp/quality behavior and prevention of misleading display
Invalid remote command	Ownership rejection and audit/diagnostic indication
Historian gap	Detection, explanation and honest representation on trend/report

Project 3 - verification and marking

Project-specific marking rubric

Category	Weight	What earns strong marks
Operator requirements/design	15%	Hierarchy, faceplates and interactions support defined operator tasks.
SCADA implementation	25%	Tags, views, commands, alarms, trends and quality behavior are robust.
Historian/alarm management	15%	Data and events are stored, retrieved and presented meaningfully.
Network/protocol engineering	15%	Topology, addressing, mappings and diagnostics are technically coherent.
Fault tests	20%	Six communication/data faults are isolated by layer and recovered.
Documentation/defence	10%	Configuration and documents agree; design tradeoffs are explained.

RELEASE STANDARD

Minimum 75/100, both prior areas supervised, all six mandatory faults closed, and no unresolved command-ownership, bad-quality or misleading stale-data defect.

Examiner questions

- [] Which control decisions remain in the PLC, and why?
- [] How does the display distinguish bad, stale and healthy data?
- [] Why can ping succeed while an OPC/Modbus connection fails?
- [] How were alarm priority and historian sampling decisions justified?
- [] What evidence isolates a mapping error from a process error?

Project 4 - Integrated Remote/Offshore Process Package Simulation

RECOMMENDED WINDOW	SYLLABUS ALIGNMENT	SUBMISSION TYPE
25 January - 28 February 2027	M7 Engineering Documentation + M8 Safety, Commissioning & Troubleshooting; integrates M1-M6	Individual industrial simulation project

PROJECT SCENARIO

A remote process package receives feed, operates pump and motor equipment, controls a process vessel, represents separator/filter duty and transfers product to storage. The learner must commission, operate, diagnose and hand over the complete simulated system.

REPRESENTATION BOUNDARY

This is an industrially inspired software simulation. It is not an oil rig, a certified offshore system, a hazardous-area design or proof of site authorization. Safety and ESD behavior is assessed only as control-system logic and documentation.

Assessment purpose

Demonstrate integrated junior-technician competence across PLCs, instrumentation, process control, SCADA, networking, engineering documentation, FAT/SAT-style testing and structured troubleshooting.

Required integrated software

Layer	Expected platform
PLC/control	TIA Portal + S7-PLCSIM or declared IEC equivalent
Process	Factory I/O or credible dynamic process model
HMI/SCADA/historian	Ignition Perspective/Maker or equivalent industrial platform
Protocols/network	OPC UA/Modbus simulation, Wireshark and OS diagnostics
Documents	Controlled drawing, spreadsheet and report tools

Project 4 - system and control requirements

The flagship must reuse and extend selected controller, process, HMI, alarm, historian and diagnostic patterns from Projects 1-3.

PROCESS/CONTROL SCOPE	PROTECTIVE/DIAGNOSTIC SCOPE
☐ Feed and storage boundaries	☐ Start permissives
☐ Duty/standby pump service	☐ Operating interlocks
☐ Motor/conveyor or transfer equipment	☐ Alarm and trip priorities
☐ Process vessel instrumentation	☐ Training E-stop/ESD logic
☐ Valve/final-element behavior	☐ First-out/cause retention
☐ Separator/filter representation	☐ Bad-value validation
☐ Manual/Automatic sequences	☐ Communications fallback
☐ Defined disturbances and limits	☐ Controlled reset and restart

Operational requirements

- ☐ Provide a defined startup, normal operation, controlled shutdown, trip and recovery sequence.
- ☐ Make command source, equipment availability and unmet conditions visible at PLC and HMI levels.
- ☐ Keep protective logic independent enough that a routine controller or HMI defect cannot silently bypass it.
- ☐ Represent instrument quality, limits, alarm/trip deadbands and final-element feedback.
- ☐ Preserve diagnostic cause through a fault cascade and require verified healthy conditions before reset.

CHALLENGE

Create at least two interacting fault scenarios in which the first symptom is not the root cause. The diagnosis must use chronology, signal traces, trends and communication evidence rather than trial-and-error changes.

Project 4 - engineering documentation package

DESIGN AND CONTROL	OPERATIONS AND COMMISSIONING
☐ System requirements specification	☐ Alarm list and response
☐ P&ID	☐ Network diagram/address schedule
☐ Electrical/control schematic	☐ Protocol/tag map
☐ PLC I/O list	☐ FAT/SAT-style procedure
☐ Instrument list	☐ I/O and loop-check sheets
☐ Loop diagrams	☐ Commissioning plan
☐ Control narrative	☐ Punch list
☐ Sequence description	☐ Troubleshooting reports
☐ Cause/effect matrix	☐ As-built change register
☐ Trip/interlock matrix	☐ Backup and handover index

Document coordination checks

Cross-check	Acceptance question
Tag	Does the same identifier appear on P&ID, I/O, logic, HMI, alarm and test records?
Range/unit	Do instrument, PLC scaling, HMI and alarm settings agree?
Cause/effect	Does implemented logic and tested response match the approved matrix?
Network	Do addresses, protocols, mappings and actual connections agree?
Revision	Can every as-built change be traced to reason, evidence and retest?

Project 4 - commissioning and acceptance

Treat commissioning as a controlled verification programme. The simulation does not remove the need for prerequisites, hold points, expected results, defect control and restoration checks.

Stage	Required evidence
Readiness review	Approved scope, architecture, software versions, document baseline and open risks
Visual/configuration check	Tag/range/address/module/connection consistency before execution
I/O checks	Stimulus -> PLC input -> logic; command -> output -> feedback for every critical point
Loop checks	Process model -> signal/raw/scaled value -> HMI/alarm/trend
FAT	Requirement-based normal, boundary, fault, recovery and sequence tests
SAT-style integration	PLC/process/SCADA/network start-up, failure and restoration in the final environment
Punch close-out	Priority, owner, correction, retest evidence, status and residual limitation
Handover	As-built files, backups, evidence index, known limitations and demonstration

Acceptance conditions

- [] Every critical requirement has a passed test or explicit NOT TESTED disposition.
- [] No open simulated E-stop/ESD, trip, restart, ownership or bad-quality defect remains.
- [] All punch items have severity, owner, status and verification evidence.
- [] A clean backup can restore the PLC, process model and SCADA configuration.
- [] The final demonstration can be completed in 5-10 minutes without hiding limitations.

Project 4 - structured fault campaign

Introduce at least ten repeatable faults across the layers below. At least two must be interacting or cascading scenarios.

Layer	Representative faults
Electrical/control representation	Open healthy chain, unavailable output supply state, feedback discrepancy
Digital I/O	Stuck input, inverted fail-safe state, incorrect address or mapping
Analog/instrument	Disconnected, biased, frozen, noisy, under-range, over-range or wrong scaling
Equipment/process	Pump failure, valve failure, no-flow, high-high level or filter restriction
PLC/sequence	Missing permissive, incorrect transition, retained state or reset defect
HMI/SCADA	Invalid command, stale value, misleading state or alarm configuration defect
Network/protocol	Connection loss, wrong IP/subnet, wrong register/datatype or service failure
Historian/time	Data gap, timestamp inconsistency or unsuitable sampling/deadband

Required report chain for every fault

TROUBLESHOOTING METHOD

Symptom -> immediate simulated process protection -> initial checks -> evidence -> isolation boundary -> diagnosis -> root cause -> controlled correction -> verification -> regression -> documentation.

Fault-report minimum

- [] Timestamped chronology and operating state
- [] Observed PLC/HMI/process/network evidence
- [] Tests considered and why they were ordered
- [] Verified root cause and rejected alternatives
- [] Correction authorization/change record in the simulation context
- [] Normal, fault, recovery and regression outcomes

Project 4 - verification and marking

Project-specific marking rubric

Category	Weight	What earns strong marks
Requirements/architecture	15%	The integrated package is coherent, bounded and traceable across all layers.
PLC/process implementation	20%	Sequences, modes, analog control and protective behavior operate correctly.
HMI/SCADA/network	15%	Supervision, quality, historian and communications support operation and diagnosis.
Commissioning evidence	20%	I/O, loops, FAT/SAT-style tests, punch control and restoration are disciplined.
Troubleshooting	20%	Ten faults, including two interacting cases, are isolated and verified systematically.
Documentation/defence	10%	As-built records agree and the learner defends decisions and limitations.

RELEASE STANDARD

Minimum 80/100 for flagship release, no category below 65%, all critical tests passed or honestly dispositioned, ten fault reports closed and no unresolved protective, restart, ownership or data-quality defect.

Five-to-ten-minute demonstration

Stage	Demonstration requirement
Scope	State process purpose, software architecture and simulation limitations.
Normal operation	Run a meaningful sequence and show PLC, HMI and trends.
Protection	Show one permissive/interlock/trip preventing an invalid action.
Fault	Introduce one representative fault and diagnose it from evidence.
Recovery	Correct, verify, regress and show the updated record.
Close	Present remaining limitations, backups and next improvement.

Portfolio final release and recruiter presentation

Present the portfolio as four coordinated industrial simulations, with Project 4 demonstrating integration and commissioning discipline.

FINAL QUALITY CHECK	INTERVIEW EVIDENCE
☐ Exactly four projects	☐ One architecture diagram
☐ All work software-simulated	☐ One PLC routine explained
☐ Each project maps to syllabus modules	☐ One analog calculation
☐ Requirements trace to tests	☐ One HMI/alarm/trend example
☐ Normal/fault/recovery proven	☐ One network diagnosis
☐ Documents and tags agree	☐ One commissioning extract
☐ Sources and exports included	☐ One fault report
☐ Limitations stated honestly	☐ Flagship demonstration video

Official software reference starting points - verified 14 September 2026

Platform	Official reference
Siemens S7-PLCSIM	https://cache.industry.siemens.com/dl/files/161/109997161/att_1348284/v1/S7-PLCSIM_Help_en-US.pdf
Factory I/O + Siemens	https://docs.factoryio.com/tutorials/siemens/
Ignition Maker	https://inductiveautomation.com/ignition/maker-edition
Ignition modules	https://docs.inductiveautomation.com/docs/8.3/getting-started/modules-overview
Wireshark guide	https://www.wireshark.org/download/docs/Wireshark%20User%27s%20Guide.pdf

REQUIRED PORTFOLIO STATEMENT

These are personal software simulation projects created to demonstrate junior automation and controls engineering reasoning. They were not commissioned on a live industrial plant and do not represent certified safety, hazardous-area, offshore or site experience.